

ARCHITECTURE OVERVIEW

The Inaya Ecosystem, End to End

How the storage protocol, the applications built on it, and the AI layer all fit together — the shape of the system, without the code.

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SECTION 01

The Shape Of The System

Inaya is one ecosystem built on two connected layers. The bottom layer is a real DePIN protocol — people encrypt their own files, independent operators store and verify them, and a blockchain anchors who owns what. The top layer is a set of applications — web, mobile, two desktop apps, and a business SaaS product — that most users interact with without ever knowing a blockchain is involved.

THE PROTOCOL LAYER

Smart contracts that handle ownership, staking, node payouts, and threat verification. This is where the token economics live, and where independent node operators actually earn.

THE APPLICATION LAYER

Everything a person actually opens: the storage dApp, the Business Workspace for companies (including its own operations, finance, and HR tooling), the mobile app, two desktop apps, a public security page, an investor data room, and four AI assistants — all talking to the same backend.

SECTION 02

Storage — How A File Actually Gets Protected

This is the core promise: nobody but the file's owner can ever read it — not Inaya, not the node storing it, nobody.

1. **Encrypt on your device.** Before a file ever leaves the user's browser or phone, it's encrypted using a key derived from their own passkey. That passkey is never sent anywhere.
2. **Split it in two.** The encrypted file is split into two halves. Neither half means anything on its own — you need both to reconstruct anything.
3. **Store both halves.** The two halves get pinned to decentralized storage (IPFS/Pinata), independent of each other.
4. **Anchor ownership on-chain.** A record — who owns this file, where its two halves live, a fingerprint proving it hasn't been tampered with — gets written to the blockchain. This is the permanent, tamper-evident receipt.
5. **Reverse it to read.** Downloading does the same thing backwards: look up the on-chain record, fetch both halves, put them back together, decrypt with the owner's passkey. Nothing about this process gives Inaya, a node operator, or anyone else a way to see the plaintext.

You don't have to take Inaya's word for any of this. The encryption code above is published as open source (@inaya-network/custody-sdk on npm) and used identically by the web app and the mobile app — not a separate, unverifiable implementation behind the scenes. Every release publishes a cryptographic hash anyone can independently reproduce and check, plus a second copy pinned to IPFS so the code itself, not just Inaya's word, identifies what you're running. Full verification steps are public.

The two halves don't depend on one company staying up. Each file's two encrypted shards are replicated across two independent pinning providers on genuinely different infrastructure (Pinata and Filebase), not just one. The system checks pin health continuously and, if a replica goes missing or its content stops matching what was originally captured, automatically fetches from the surviving healthy copy, re-verifies it, and re-pins — self-healing back to fully protected without anyone having to notice or act.

SECTION 03

Node Operators — Who Actually Stores The Data

Independent people and organizations run node software that registers them on-chain and reports their capacity. In exchange, they earn USDT and \$INAYA — for storage service, and separately, for contributing to a decentralized threat-intelligence network (see Section 05).

- Registration and identity live on-chain — a node's stake, tier, and track record are public and verifiable.
- Payouts are deliberately delayed and verifier-attested rather than instant and self-reported — this protects the system from a single bad actor (or a single compromised key) draining funds in one move.
- Today, the software an operator runs handles registration and reporting — it doesn't yet handle the actual physical hosting of file shards. That's a known, stated gap in the current build, not a hidden one.

SECTION 04

The Web dApp — Where Consumers And The Protocol Meet

The main website is where an individual user does everything token-and-storage related: gets testnet tokens from a faucet, uploads/downloads files in the Sovereign Vault, stakes \$INAYA for yield, tracks their activity on a dashboard, refers friends (email + identity verification, no wallet needed for that part), and — for larger customers — buys an annual Corporate Reserve storage plan, paid in USDT and settled on-chain.

Staking has real lock-period tiers (flexible, 30-day, 90-day) with better rewards for longer commitments — this is enforced on-chain, not just a UI label.

SECTION 05

Business Workspace — A Second Product On The Same Foundation

Not every customer wants a blockchain product. Business Workspace is a genuine B2B SaaS tool — organizations, departments, projects, documents — with real approval workflows, granular permissions, secure sharing links, and a permission-aware AI assistant. Sign-in is just an email or a Google account. No wallet, no tokens, no blockchain vocabulary anywhere in the experience.

It has grown well beyond documents: real business operations (Tasks, CRM, Procurement, Inventory) and a Finance & HR layer (invoicing, expense approval, employee records, leave management) now run on the exact same organization/department/permission foundation described above — every new module additive, nothing about the base system had to be restructured to add them. Finance & HR is explicitly a testnet demonstration layer today, not regulated banking or payroll.

Why this matters strategically. It's a second, independent revenue engine — priced and sold like any other business software, evaluated by a buyer who will never ask what chain it runs on — while still running on top of the same infrastructure and the same encryption guarantees as the consumer storage product.

SECTION 06

Security Layer — A Decentralized Threat Network

Marketed publicly as "Inaya Firewall." The same trust model as the storage layer, applied to threat intelligence instead of files: independent nodes report suspicious websites, the network only trusts a threat once enough independent, reputable reporters agree, and confirmed verdicts get permanently recorded on-chain so they can't quietly be changed later.

- A public website page lets anyone — no account needed — check a domain, see network stats, and ask an AI assistant about how it all works.
- The mobile app can check destinations before you trust them and keep your own personal block/allow list.
- The desktop apps can, on Windows, actually block a confirmed-malicious address at the operating-system firewall level, not just inside the app window.

SECTION 07

The AI Layer — Four Assistants, One Pattern, Different Rules

All four assistants (Docs, Business, Security, Learn Tutor) are built the same technical way — they can call small, purpose-built tools to look up real data before answering. What's different is the philosophy each one follows.

ASSISTANT	ITS ONE RULE
Docs Assistant	Answer grounded in the project's own real documentation — retrieves the relevant passage first, then answers from it, and cites where the answer came from.
Business Assistant	Never show a user data they aren't actually permitted to see — every answer respects real document/project/finance/HR permissions.
Security Assistant	Never invent a threat verdict — every specific claim must be backed by real, verified network data.
Learn AI Tutor	The opposite of the other two — teach freely using its own knowledge, like a real tutor would; only use tools to check the user's own saved videos and progress.

How the Docs, Security, and Learn assistants find the right passage. A shared retrieval layer (RAG) indexes the project's real documentation and FAQs, embeds it for semantic search, and combines that with plain keyword search — so "how does the pricing work" and "pricing structure" both find the same real answer, not a static block of hardcoded text that goes stale the moment the product changes.

AI-Powered Business Operations — the AI can propose, it can never execute.

As of September 2026, the Business Assistant can go further than answering questions: it can propose real changes across 9 business domains (Tasks, Expenses, Documents, Employees, Invoices, Leave Requests, Purchase Orders, Purchase Requests, Deals). It never performs any of them itself.

- 1. The AI proposes, not executes.** Every proposal is checked against the exact same permission the real action would require before it's even created — the AI can't propose something the requesting user isn't already allowed to do.
- 2. A human approves with real authority.** A person with the exact same real authority the underlying action requires — re-checked independently by the server, not trusted from the request — must approve it. Higher-risk actions (terminating an employee, approving a large purchase) are automatically classified as higher risk than routine ones (advancing a task).
- 3. A mandatory 36-hour delay passes.** Even after approval, nothing executes immediately — mirroring the same timelocked-settlement pattern the storage protocol itself uses for node-operator payouts, reused here for a completely different kind of action.

4. **Only then does it execute — and everything is recorded.** A separate process executes the change only once the delay has genuinely passed, and only if the real action still makes sense. Every step — proposed, approved, delayed, executed, rejected, or expired — is written into a cryptographically verifiable audit trail.

The audit trail isn't just a log — it's tamper-evident by construction. Every entry cryptographically commits to the one before it, so altering or deleting any past entry breaks every entry after it — a direct database edit is mathematically detectable, not just against policy. Any business customer can independently walk their own organization's chain and recompute it themselves via a self-service export; they don't have to trust a "Verified" badge Inaya shows them. Security-tested with 19 automated tests, 11 of them adversarial scenarios specifically trying to defeat the guardrails.

SECTION 08

Mobile & Desktop — Same Backend, Different Doors

None of the client apps are separate products with their own logic — they're different doors into the exact same backend and the exact same data.

- The mobile app is a full superset — everything the web dApp does, plus the Business Workspace, Learn, and Security, in one app.
- The two desktop apps are deliberately thin — each one is just the real website, running inside a lightweight native window with a system tray icon, native notifications, and auto-updates layered on top. There is no separate desktop-only feature set to keep in sync.

SECTION 09

App Store, NFT Vault & Account Security (September 2026)

Three smaller but real additions rounding out the consumer side: a place to discover Web3 apps, a place to back up NFTs the same way files are backed up, and a stronger login.

- Web3 App Store (/apps) — a curated set of Inaya's own apps, plus community-submitted ones. A submission needs a wallet signature and is checked against the live threat network before an admin ever sees it, and nothing goes live until approved.
- NFT Vault (/nfts) — discovers NFTs a wallet owns and backs up their metadata and image using the exact same encrypt-and-shard pipeline that protects every other file on Inaya.
- Optional two-factor login for Business Workspace — an authenticator app or a text message code, on top of the existing passwordless sign-in.

SECTION 10

Multi-Chain — Moving Value Beyond BNB Chain

Inaya started on BNB Chain Testnet, but \$INAYA doesn't have to stay there. Two independent systems, running in parallel, let value move to other chains — deliberately kept separate, since a problem in one says nothing about the other.

INAYA'S OWN NATIVE BRIDGE

Full control, no third-party dependency. A real, proven lock-and-mint cycle now works to Ethereum, Avalanche, Arbitrum, Solana, Hedera, Aptos, and Sui — each one verified with a real transaction that moved real (testnet) value and confirmed it arrived on the other side, not just deployed-and-assumed-working.

A SECOND ROUTE VIA WORMHOLE

An independent, third-party interoperability network, integrated as an alternative path that doesn't replace the native bridge. Proven working end-to-end to Ethereum, Arbitrum, and Avalanche; a few other chains are currently blocked by issues in Wormhole's own tooling, not anything on Inaya's side, and are shown as reference-only until that's resolved upstream.

Nothing here is claimed beyond what's actually been proven on-chain. A chain only ever gets credit for the capability level it's genuinely demonstrated — discovered, wallet-connected, messaging-only, or a real proven transfer — never assumed from "it's deployed" alone. Several real bugs were found and fixed getting the newer chains (Solana, Sui) working; each is documented rather than smoothed over, the same honesty convention this whole ecosystem holds itself to.

SECTION 11

Oracle & Automation — Infrastructure That Runs Itself

A third independent subsystem: an on-chain registry of approved data sources, and a keeper that executes pre-approved contract actions automatically — under smart-contract rules, never arbitrary admin commands. Live on BSC Testnet today, publicly verifiable at inayanetwork.com/automation.

- The live demo isn't simulated: the price feed reads the real PancakeSwap testnet pool for INAYA/USDT, and the automated action is a real, already-existing node-settlement release function on InayaNodeRegistry — the first run found and released an actual previously-unclaimed payout.
- If the data feeding an automated action goes stale, the system skips that action rather than acting on unverified information — proven live, not just claimed.
- The keeper never holds broad authority — it can only submit data it's explicitly registered for, and only call functions that are already safe for anyone to call.

SECTION 12

Sovereign Enterprise OS — One Operating Environment, Not a Pile of Features

Almost everything described in this document already existed as a separate feature. What was missing was the connecting layer — one place that shows you what needs your attention, lets you search everything at once, asks one assistant instead of four, and tells you plainly whether your data and your organization's activity are in a trustworthy state right now.

- One OS Home screen — trust & health status, recent notifications, what changed this week, and an assistant that can answer both business and security questions — on both the web app and Business Workspace, each tailored to how you actually use it.
- Real notifications with a memory — mark something read once, and it stays read, instead of the app forgetting the moment you refresh.
- One search bar (Cmd/Ctrl+K) instead of hunting through a dozen separate screens for a task, a deal, a document, or a file.
- A trust & health signal that's actually true, not decorative — built from the same real audit trail, backup health, and security data described earlier in this document, never invented for the sake of a clean-looking dashboard.
- On desktop, a module you're watching closely (like the audit trail) can now pop out into its own window and stay visible alongside everything else, instead of switching away from what you were doing.

Two operating environments, not one blended one. Your business identity (email, organization membership) and your wallet identity are kept genuinely separate — there was no clean way to merge them without inventing a new identity system neither product actually needed, so the OS shell was built twice from the same underlying pieces instead: one for Business Workspace, one for the wallet-connected app. This mirrors how the two desktop apps already work today.

SECTION 13

Healthcare & Legal OS — Two Industries, Same Secure Foundation

Business Workspace now has two specialized modes for organizations with genuinely different needs: Health OS for hospitals and clinics, Legal OS for law firms and legal departments. Both are built on the exact same encryption, audit trail, and permission model as the rest of Business Workspace — just adapted to the records those two industries actually work with.

- Need-to-know access, not just department-wide — only staff actually assigned to a patient's care team, or a lawyer actually on a matter's team, can see that record. Being in the right department isn't enough on its own, which is stricter than how the rest of Business Workspace works, on purpose.
- Emergency access that's actually safe — a clinician can get emergency access to a patient outside their normal assignment, but it's logged the instant it happens, expires automatically after a few hours, and a manager is notified to review it. We caught and fixed a real bug during testing where an early version didn't expire on time.
- A legal hold that actually holds — placing a hold on a matter doesn't just show a warning, it genuinely blocks anyone from deleting the related records until the hold is released.
- Trust accounting with one hard rule — the system will never let anyone withdraw more from a client's trust fund than what's actually there, no matter who's asking.
- An AI assistant that knows its limits — it can summarize, search, and draft, but it's built to flatly refuse anything that crosses a real line: diagnosing a patient, giving final legal advice, or agreeing to release a hold or delete evidence.
- The right modules for the right business — a general company never sees Health OS or Legal OS at all; a hospital or law firm picks its type at signup (or changes it later in Settings), and that choice is enforced for real, not just hidden behind a menu a determined user could still reach.

What's live today vs. what's next. Patient records and legal matters both have a real, working screen in Business Workspace today. Most of the other modules described above — consent, billing, scheduling, evidence, legal holds, contracts, trust accounting — work correctly behind the scenes and are already tested, but don't have their own screen in the app yet; that's the next phase of work, building on logic that's already written rather than starting over. Nothing here is HIPAA-certified, ABA-certified, or legally compliant with any specific jurisdiction — that would require a real legal/compliance review this build doesn't include, and we've been careful not to claim otherwise.

SECTION 14

Financial Services, Regulated Enterprise & Government — Three More Industries, Same Foundation

Three more specialized modes now sit alongside Health OS and Legal OS, all sharing the exact same encryption, audit trail, and permission model: Financial Services OS for hedge funds and asset managers, Private Capital OS for private equity and venture capital, Regulated Enterprise OS for banks, insurers, and other heavily-regulated companies, and Government OS for public-sector departments and agencies.

- A compliance foundation that never overstates itself — controls can be mapped to real industry frameworks (NIST, ISO 27001, SOC 2, DORA, GDPR), but every mapping is clearly labeled a reference, never a certification. A control nobody has tested yet shows up as 'unknown,' never a falsely reassuring green checkmark.
- A published policy can't be quietly rewritten — changing it always creates a brand-new version, so there's always a clear record of what a rule said on any given day. The Government OS Policy Knowledge Base reuses this exact same rule.
- Real fund, deal, and investor tracking for the finance side — research, investment committee decisions, portfolios, and for private equity/VC: deal pipelines, due diligence, cap tables, and portfolio-company boards.
- Government's need-to-know rule, borrowed from Health OS — a caseworker with general staff access still can't open a specific citizen's record unless they're actually assigned to it. Every citizen-record read is logged, not just every change — a stricter bar than the rest of Business Workspace.
- An outside examiner or auditor gets their own scoped, one-time-use link — never a real login to the company's system — for both regulatory examinations and Government OS's own case reviews.
- AI assistants that know their limits here too — they can search, summarize, and draft, but are built to flatly refuse a request to certify compliance, approve something final, or override a permission, on any of these four verticals.

What's live today vs. what's next. Financial Services OS, Private Capital OS, and Regulated Enterprise OS are fully built on the web app, verified end-to-end. Government OS's core (citizen records, case management, the policy knowledge base, the operations/security dashboard, and its own AI assistant) is also built and tested on web. None of the four have a mobile screen yet — that's next, building on logic that's already written. No certification, accreditation, or jurisdiction-specific regulatory approval of any kind is claimed for any of these — that requires independent authorities this build doesn't include, and we've been careful not to claim otherwise.

SECTION 15

Trust You Can Check Yourself, and Recovery That's Actually Re-Tested

Two additions that both answer the same question a different way: not 'trust us,' but 'here's how you check.'

- A public Trust Center (inayanetwork.com/trust) that anyone can visit without logging in — and its centerpiece isn't a badge that just says 'Verified.' It recomputes the cryptographic proof itself, in your own browser, and tells you exactly where a record breaks if it ever does.
- We found a real bug in that very verifier while testing it against a real record: it originally assumed every proof started from the very beginning of the chain, so a smaller, single-record proof was wrongly flagged as broken. Fixed, and now the page is honest about which of the two proof strengths you're actually looking at.
- A real way for two organizations to formally trust each other — propose, accept, revoke — though today it's the foundation for that, not yet a feature anything else in the app uses to actually share data.
- Developer API keys that always act as the organization that created them, never anyone else's — down to the design, not just the policy.
- Recovery testing that actually runs, automatically, against real systems — using safe stand-in files Inaya owns itself, since testing with a real customer's files would require a password Inaya was never given in the first place. Every test's real pass/fail result becomes part of the same tamper-evident record everything else in this document already uses.

What this doesn't claim. No SOC 2, HIPAA, ABA, or FedRAMP certification is claimed on the Trust Center — it says so directly. And a specific customer's own resilience-test results stay private to that customer, never published on the public page.

SECTION 16

A Real Home for Node Operators

The people who actually run Inaya's storage nodes now have their own dashboard — separate login, separate from the wallet app, separate from Business Workspace — showing uptime, rewards, and network standing.

- Real uptime history, not guessed — and if the dashboard doesn't yet have enough history to answer a question honestly (say, 90 days of data on day one), it says so instead of estimating.
- Real numbers read directly from the blockchain for the things that matter most — an operator's tier and earnings — not a copy that could go stale.
- There's deliberately no 'release my payment' button — payouts are already fully automatic on a timer nothing in this dashboard can shortcut; it can only show you where things stand.

SECTION 17

Four New Ways to Prove and Protect Business Value

Four new tools in Business Workspace, each doing something genuinely useful — described precisely, so nobody mistakes what's cryptographically guaranteed for what's a well-built internal workflow.

- Inaya Sign — document signing with a real digital signature bound to the exact document and moment it was signed; if the document changes afterward, that signature stops being valid for it. A completed signing can optionally be stamped onto the blockchain as extra proof, on top of the signature itself.
- DePIN Storage Manager — lets a business register and monitor its own storage nodes today; it doesn't yet route actual files through them the way the public node network does, and we say so plainly in the product rather than implying it already does.
- Milestone Escrow — a real, enforced 'don't release payment until a milestone is confirmed' workflow, with mandatory human approval and a built-in waiting period before money actually moves. It's an internal safeguard, not a blockchain escrow contract holding funds — we describe it that way on purpose.
- Financial Attestation — a business can cryptographically prove a claim about its own finances (like 'revenue exceeded \$X this quarter') without a third party needing to see every underlying transaction, using a real cryptographic fingerprint that changes if a single number is later altered. It's a strong, real guarantee — just not the specific 'zero-knowledge proof' technique, which this doesn't use.

SECTION 18

Smarter Day-to-Day Operations

Real improvements to the daily business tools, plus an AI assistant that guides rather than acts.

- Invoices and purchase orders now support five currencies, always showing the original amount and a converted figure side by side, using a clearly dated reference rate rather than a live market feed.
- One-click professional invoices, moving stock between warehouses with a full paper trail, editable purchase orders before they ship, and new downloadable reports.
- An AI assistant that can now walk someone through a multi-step task, one real step at a time — but it never performs the step for you. It only advances once it sees real proof you actually did it, not just because you told it you did.

SECTION 19

Real Connections to the Tools You Already Use — and a Voice to Talk To

Two upgrades to how people actually interact with the platform day to day.

- Nine business integrations (accounting, communication, storage, and productivity tools) now connect for real — a genuine sign-in with each provider, not a placeholder 'Connect' button. Disconnecting one now genuinely revokes access, instead of just hiding a status.
- The Business AI Assistant can now be spoken to and talks back — the same assistant, the same limits, the same inability to take a real action without a real person's approval. Voice is a new way to reach it, not a new set of permissions.

SECTION 20

A Workspace That's Easier to Actually Use

A real usability pass across Business Workspace — not a visual redesign, since the look was already consistent. The actual problem: the same interface pieces had been hand-copied dozens of times instead of built once and shared, and a few navigation choices genuinely confused people.

- Two competing 'home' screens became one. Clicking Departments, Projects, or Documents used to all land on a screen generically titled 'Company Records' no matter which you clicked — now it correctly shows which one you're in.
- Every form field got a real, readable label for the first time — previously, faint placeholder text was the only clue what a field was for.
- None of this touched how permissions, security, or any backend logic works — it's exclusively about making the same real functionality easier to find and use.

SECTION 21

Multi-Cloud Storage Compatibility & Enterprise Storage Governance (September 2026)

The storage layer now speaks the two protocols enterprise IT already standardizes on — a real AWS S3-compatible API and a real Azure Blob Storage-compatible API, each verified against the actual AWS CLI and the official Azure SDK, not a simplified approximation. An organization's existing scripts, backup tools, and SDKs connect directly, with no migration and no new tooling to install.

- Scoped access credentials — a credential can be restricted to one bucket, one folder path, a specific set of operations (read/write/delete/list), and an expiry date, enforced server-side on every request.
- Object versioning — overwriting a file keeps prior versions retrievable and restorable, opt-in per bucket.
- Object Lock — immutable, time-locked retention for compliance, enforced identically whether the request comes through the S3 API, the Azure API, or Business Workspace.
- Legal hold — an indefinite deletion block for litigation/investigation holds, independent of any retention timer.
- Lifecycle policies — automated expiration by age, with locked or legally-held objects always exempt.
- Inaya Drive — a real Windows drive letter backed by the same storage, live-tested end to end: mount, browse, open, save, and delete files the same way you would on any local or network drive.

How this fits the zero-knowledge model described in Section 02. Standard S3/Azure client tools were never built to encrypt a file client-side before sending it — that step doesn't exist in the AWS CLI or the Azure SDK. So objects written through this compatibility layer specifically use a server-managed encryption key, scoped per organization, itself encrypted at rest, with every use logged to that organization's own audit trail — the same trust model any real S3-compatible gateway product operates under. This is a disclosed, structurally necessary exception for this one access path; every other Inaya document — uploaded through the web app, mobile app, or Business Workspace — keeps the client-side-only model Section 02 describes, unchanged.

Inaya Drive's WinFSP-backed filesystem code runs as its own separate process (inaya-drive-helper), not linked into the desktop app's own binary — a deliberate licensing boundary, since the real Rust WinFSP binding is GPL-3.0 and this keeps that license off Inaya's proprietary desktop app entirely.

SECTION 22

How It All Actually Connects

One backend serves every surface — the website, the mobile app, and both desktop apps all call the exact same API. There's no duplicated logic to keep in sync between platforms; a fix or feature on the backend is instantly live everywhere.

The protocol (contracts, node operators, encryption) and the applications (dApp, Business Workspace, mobile, desktop, AI) are cleanly separated — most of what a user touches day-to-day never has to think about the blockchain underneath it, even though it's doing real work a layer down.

BNB Chain Testnet is still home base — the core protocol, staking, and node settlement all live there — but it's no longer the only chain involved: \$INAYA now moves to several other testnets through the bridge described in Section 10. Nothing described in this document is live on mainnet yet, on any chain — that's the next major milestone, not a past one.